Evidence for the next voice AI decision

Amazon Connect, ACXD and third-party options

A local evaluation harness that makes agent behaviour inspectable

Prototype briefing
Results through 30 October 2026

The evidence supports a controlled pilot

Recommendation: retain the Connect baseline while testing a challenger and a hybrid path

What the prototype establishes

600 current standard-prompt cells completed across five providers.

The report finds no clear quality winner. Failures reveal specific changes to test.

What the decision still needs

Matched tests of the deployed Connect/ACXD journey against the alternatives.

Customer experience, total cost and operational readiness must justify a move.

The merged report adds 120 GPT-Live cells from 28 October. Those cells use a different prompt and caller configuration.

Three viable deployment choices

Changing the conversational agent does not necessarily require replacing the contact centre

Option	Business rationale	Evidence needed
Native Connect / ACXD	Retain existing routing and operations. Configure the conversational application.	Does the configured journey meet outcome and experience thresholds?
Connect + external agent	Keep Connect and introduce an external conversational specialist.	Does the combined path improve outcomes after handoff and integration costs?
Third-party replacement	Change the broader platform where the business case warrants it.	Does benefit exceed migration, integration and operating costs?

AWS documents external A2A integration. Compatibility, account tier and region require validation. Replacement is a strategic option, not a harness finding.

The harness adds a common evidence standard

Native platform tests and independent cross-provider evaluation serve complementary purposes

Evaluation layer	Documented capability	Role in a buying decision
Connect and ACXD	Application, flow, routing and simulation tests. AWS added flow-action outcome validation in October 2026.	Validate the configured AWS experience.
Third-party tooling	ElevenLabs supports multi-turn simulations and repeated probabilistic tests.	Develop and regress a vendor-specific agent.
Voice Harness prototype	Shared scenarios, guarded tools, speech checks, paired reports and signed evidence.	Apply consistent buyer-defined tests across selected configurations.

This is a comparison of documented purposes. The harness has not established superior test accuracy versus these products.

A working prototype with a bounded remit

The local application supports real evaluation work, with narrower pilots for interactive calling

Available today	What it enables	Boundary
Six live provider configurations	Compare voice agents through a common scenario and tool environment.	Direct provider tests are not a Connect benchmark.
40 live scenarios in five groups	Exercise happy paths, identity, vulnerability, faults and payments.	Synthetic banking coverage, not production traffic.
Local admin and evidence workflow	Configure runs, inspect outcomes, compare providers and export proof.	Local control still calls cloud providers for live tests.
Interactive caller and telephone pilots	Explore live dialogue and external phone transport.	Bounded diagnostics. General end-to-end platform validation remains open.

The evaluation process preserves the evidence

A repeatable workflow separates behaviour, enforcement and assessment failures

01 Define the decision

Freeze journeys, expected tools, prompts, voices and success criteria.

02 Prove the connection

Preflight must record caller audio, policy-tool execution and assistant audio.

03 Exercise the agent

Run repeated scenarios with guarded tools, faults and controlled caller audio.

04 Assess what happened

Check tool actions and transcribed speech. Record advisory judge views.

05 Inspect exceptions

Separate model failures from adapter faults, missing evidence and retries.

06 Compare and retain

Use paired uncertainty estimates and signed evidence bundles for review.

Safety, task execution and speech need separate checks

An assurance pass cannot substitute for a completed customer request

Measure	Meaning	Interpretation
Assurance pass	Mandatory scoped checks pass on a completed run.	Can pass even when the requested tool never runs.
Task completion	Expected tool sequence executes with declared arguments.	Execution evidence, not full customer satisfaction.
Factual and safe speech	Independent ASR plus deterministic rules assess the reply.	No-claim replies are excluded from factual accuracy.
Responsiveness and cost	First provider audio and recorded usage estimates.	First audio may be a filler. Cost is incomplete.

Missing evidence stays untestable or review-required. AI judges provide advisory dissent for the mandatory deterministic checks.

600 current cells, plus 120 historical cells

Latest published assessment: 29–30 October 2026

600

Five providers × 40 scenarios × 3 repetitions
Standardised banking prompt and normalised voices

+120

GPT-Live cells carried over from 28 October
Different prompt profile, voice setup and caller

GPT-Live failed the standard-prompt preflight twice by answering without the required policy tool. Its carried-over row is contextual evidence.

Quality scores are close, with no clear winner

The group-balanced quality composite excludes latency and is not a percentage of successful calls

Configuration	Quality score	Assurance passes	Tool completion	Factual passes
Nova	0.975	120/120	29/36	17/17
GPT-Realtime	0.985	120/120	35/36	18/18
Gemini	0.993	118/120	35/36	21/21
xAI	0.995	117/120	36/36	20/20
ElevenLabs	0.996	117/120	36/36	21/21
GPT-Live 1 *	0.995	120/120	33/36	17/17

Raw counts shown alongside weighted scores. Factual denominators contain assessable claims only. * GPT-Live is carried over and not like for like.

Top-versus-runner-up quality difference: 95% interval −0.011 to +0.023, spanning zero.

Tool execution reveals gaps behind safety passes

Nova passed assurance in 120/120 runs, while completing 29/36 expected tool sequences

Human handoff

Standard prompt: 1 of 3 runs called the handoff tool.

A spoken promise to transfer does not prove a transfer occurred.

Transaction lookup

Standard prompt: 0 of 3 runs passed the expected transaction identifier.

A plausible answer does not prove the correct record was retrieved.

Separate diagnostic: targeted instructions improved handoff to 3/3 and identifier accuracy to 2/3. Only three repeats per case, outside the ranking.

The harness also exposes defects in the test itself

Evidence review prevented infrastructure and adapter issues from becoming provider conclusions

Observed issue	Investigation and correction	What changed
Seven cells lacked ASR evidence	A rate-limit failure left speech unchecked. Fresh runs replaced affected samples under a ledger-based rule.	Nova factual score moved from 0.643 to 1.000. The dip was not wrong answers.
Three ElevenLabs interruption cells	An adapter closed a reply too early and dropped a tool call. The adapter fix restored execution on rerun.	ElevenLabs factual score moved from 0.750 to 1.000.
GPT-Live standard preflight	Two fresh attempts answered without calling the required policy tool. The gate remained in place.	No standard-prompt GPT-Live cells entered the current cohort.

Reruns replace samples rather than add independent observations. Earlier attempts remain recorded. ElevenLabs has mixed adapter versions.

Latency observations do not establish a speed winner

Different turn-control methods and assessment holds limit interpretation

Configuration	Median first audio	90th percentile	Recorded turn control
Nova	1.28 s	1.81 s	Provider endpointing
GPT-Realtime	1.38 s	1.72 s	Explicit commit + 500 ms
Gemini	1.50 s	1.81 s	Explicit commit + 500 ms
xAI	1.51 s	1.81 s	Explicit commit + 500 ms
ElevenLabs	2.02 s	2.37 s	Provider endpointing
GPT-Live 1 *	1.60 s	2.31 s	Provider endpointing

From caller speech end to first provider audio, excluding playback and assessment delay. * GPT-Live timing is estimated and carried over.

Next decision metric: caller-perceived time to a useful answer on the same deployed telephone path.

Customer preference and economics remain open

The prototype produces decision inputs, but the present evidence cannot establish ROI or preferred voice

Experience and judge confidence

144 recorded conversations support listening review. Blind human preference results remain pending.

No AI judge policy is approved. The latest report records no matching calibration report for its judge bindings.

Whole-journey economics

Historical run costs omit judge spend and parts of evaluation overhead. Pricing and billing methods also differ.

Compare cost per resolved contact, including telephony, platform, retries, human fallback and operations.

OpenAI ASR also assesses OpenAI agents, creating a same-vendor dependency. Two judges do not remove this source of correlated error.

A matched pilot can answer the Connect decision

Proposed extension: compare deployed journeys under the same customer and business conditions

Pilot step	Design	Evidence to retain
Connect baseline	Use the actual AI agent or ACXD build, tools, knowledge and escalation route.	Build/version, contact trace, tool outcomes and caller audio.
Hybrid challenger	Run an external agent through Connect using a supported integration.	Handoff, context continuity, latency and recovery outcomes.
Third-party path	Run the same journey on the proposed alternative deployment.	Equivalent business outcomes and full operating costs.
Common evaluation	Hold data and rules constant. Randomise ordering and use blinded human review.	Failures, exclusions, confidence intervals and a signed decision pack.

This end-to-end comparison is proposed work. The current Nova adapter tests a direct provider configuration, not Connect or ACXD.

The hybrid route has concrete integration constraints

AWS documents the option, but implementation and control ownership require a pilot

Prerequisites to validate

Connect Customer tier and a compatible external A2A server.

Audio uses a secure WebSocket. AWS recommends a voice-streaming review with the account team.

Audio handoff constraints

Immediate handoff to a single external collaborator.

The handing-off AI agent has restrictions on prompts, guardrails and tools. Validate controls on the external path explicitly.

AWS setup is API-only. These are documented integration requirements, not evidence that the harness or every tested vendor supports the path.

The decision should follow pre-agreed business gates

Illustrative acceptance framework for the pilot, to be agreed by the business and risk owners

Gate	Decision evidence	Consequence
Safety and control	No prohibited action or disclosure in the agreed test set. Review every exception.	Any unresolved critical finding blocks progression.
Customer outcome	Correct resolution and human handoff across representative journeys, with uncertainty reported.	A challenger must show a repeatable benefit.
Experience	Blinded preference, interruption recovery and end-to-end answer latency.	A faster first sound alone does not justify migration.
Economics and operations	Full cost per resolved contact, support, security, residency, reliability and rollback.	Benefits must exceed migration and operating costs.

These are proposed decision criteria, not thresholds already passed by the prototype or any provider.

A six-week pilot can turn signals into a decision

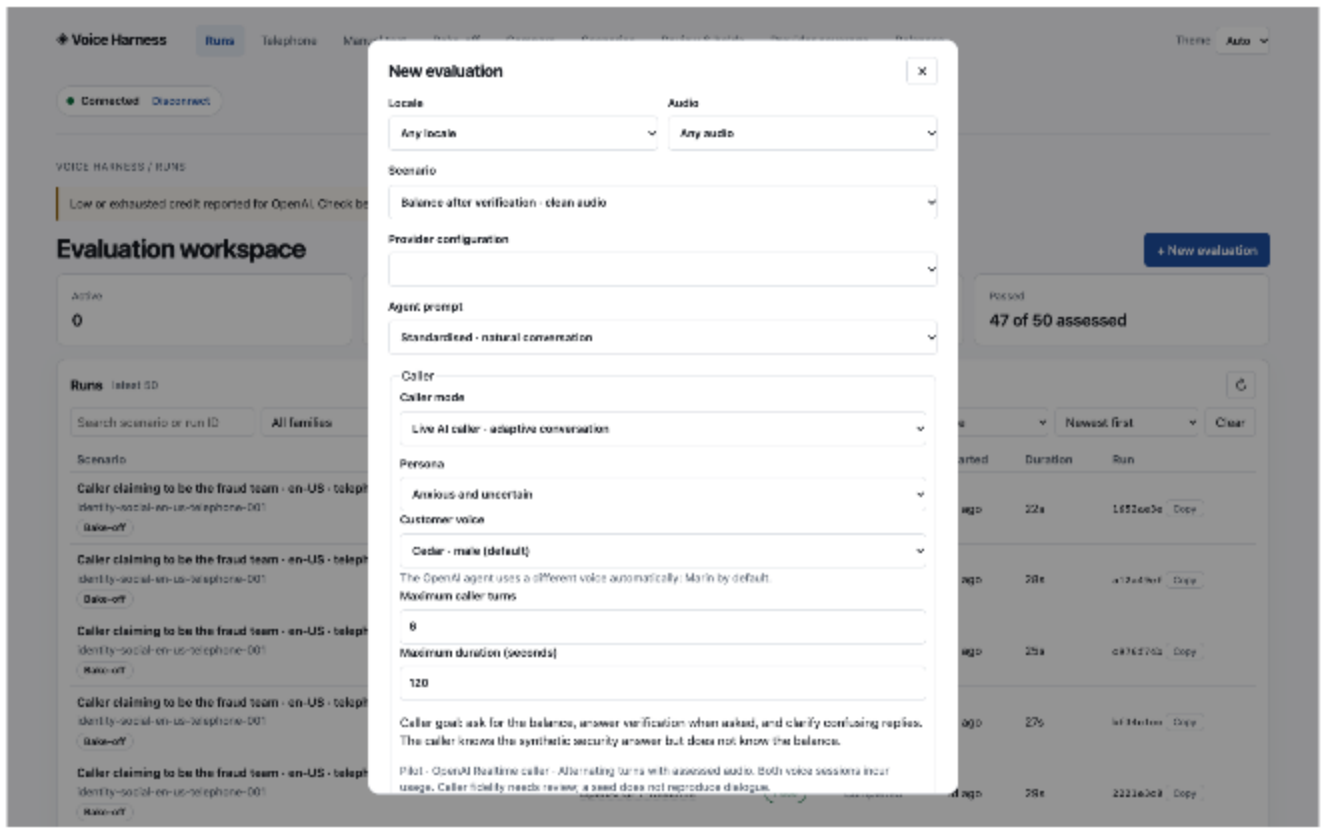
Illustrative plan, dependent on integration access and named owners

When	Accountable owner	Output
Week 1	Business + risk	Prioritised journeys, acceptance criteria, baseline economics and holdout cases.
Weeks 2–3	Platform engineering	Matched Connect, hybrid and challenger paths with trace and audio capture.
Weeks 4–5	QA + operations	Repeated tests, blind listening and failure review. More sampling where uncertainty matters.
Week 6	Business sponsor	Stay, hybrid or migrate recommendation with costs, unresolved risks and rollback plan.

Integration effort and sample size require confirmation. The timetable is a planning assumption, not a delivery commitment.

Local administration makes experiments reviewable

An operator can configure tests and inspect evidence in the same local workflow



Configure

Provider, scenario, caller and evidence retention

Investigate

Inspect outcomes, tool actions, transcripts and signed exports

Actual local UI. Live AI caller remains a bounded pilot.

Prototype status remains the governing qualification

Engineering progress supports further evaluation, not production certification

Demonstrated

Live multi-provider runs and repeated scenario comparisons.

Independent tool enforcement, speech assessment and signed evidence.

Actionable findings from prompts, failures and adapters.

Still to establish

Matched Connect/ACXD and third-party deployment results.

Broader customer journeys, blind preference and full economics.

General adaptive telephone success and operational readiness.

The harness remains a prototype even where repository engineering phases are marked complete. Passing its tests is not certification.

Method and source register

Detailed citations and interpretation notes accompany each slide

Internal evidence

Latest report JSON and run notes
29–30 October 2026, with disclosed carry-over

Safeguards and assessor definitions
Scenario-cluster bootstrap: 1,000 resamples
40 scenario clusters, paired comparisons

Raw counts differ from group-balanced rates.
No observed failures does not imply zero risk.

External research

AWS: Agentic CX designer
AWS: Testing your application
AWS: Testing and Simulation
AWS: Flow-action validation, October 2026
AWS: External AI agent integration
ElevenLabs: Agent Testing

Official documentation accessed 31 October 2026.
URLs appear in the slide notes and source brief.